

Password Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacks and Exploits

Objective: 4.3: Given a scenario, perform authentication attacks using the appropriate tools.

Video Overview

This video will provide a thorough understanding of password attacks. You will learn about various methods used to compromise passwords and gain unauthorized access to systems, from hash cracking to more advanced techniques like Kerberoasting and NTLM Relay Attacks. This knowledge is essential for aspiring cybersecurity professionals as it equips you with the insights needed to both identify and defend against such vulnerabilities in real-world scenarios, strengthening your ability to protect sensitive information and maintain system integrity in your future careers.

Key Topics Covered

- **Automating Password Discovery:** Focusing on grabbing and cracking hashed passwords to reveal original credentials for mimicking valid users through pass-the-hash attacks.
- **Hash-Based Attack Techniques:** Primary methods include brute-forcing (trying every character combination) and dictionary attacks (using pre-compiled wordlists).
- **Word List Generation:** Tools like Cewl for custom wordlists by scraping websites and pre-built lists such as Rockyou.txt and SecLists for common and leaked credentials.
- **Hash Cracking Tools:** John the Ripper and Hashcat as powerful tools for performing hash cracking operations, often in post-exploitation scenarios.
- **Password Spraying (Credential Stuffing):** Trying a few common passwords across many user accounts to exploit password reuse vulnerabilities and avoid account lockouts.
- **Password Spraying Tools:** Hydra and Medusa for feeding lists of usernames and passwords to target various services like FTP.
- **Kerberoasting:** An advanced Windows environment attack targeting Kerberos authentication protocol, exploiting Service Principal Names

(SPNs) for lateral movement and privilege escalation by capturing service account password hashes.

- **NTLM Relay Attacks:** On-path attacks for Windows systems relying on NTLM authentication, involving poisoning network services (LLMNR and NBT-NS) to intercept and relay authentication requests and capture NTLMv2 hashes.
- **NTLM Relay Tool:** Responder as a common and effective tool for NTLM relay attacks in Windows environments.

Student Notes Section